

EXP : 15

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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Naive Bayes classifier
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Display predictions
print("Actual Values:")
print(y_test)
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print("\nPredicted Values:")
print(y_pred)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")

# Display confusion matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

# Display class names
print("\nClass Names:")
print(iris.target_names)

... Actual Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy: 97.77777777777777 %

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

Class Names:
['setosa' 'versicolor' 'virginica']

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